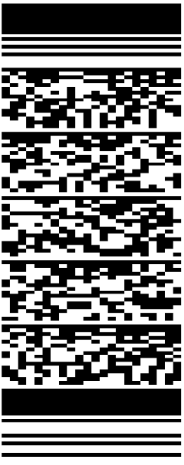


VISION Resistivity		Schlumberger	
Dual Frequency			
Recorded Mode, Run 12			
Company:	Equinor		
Well:	31/5-7		
Field:	Eos		
Rig Name:	West Hercules		
Country:	Norway		
Latitude:	60° 34' 35.14" N	UWID:	
Longitude:	3° 26' 36.13" E	Rig Name:	West Hercules
Block:	31/5	Rig Type:	Semi-Submersible
FL:	Norwegian North Sea		
FL1:	Section: 8 1/2 in.		
FL2:	Job no.: 19SCA0085		
Log Measured From: - Drill Floor: 31.00 m Permanent Datum: - Mean Sea Level Ground Level: 307.00 m Other Services: Direction & Inclination (Surveys) Annular Pressure While Drilling geoVISION Bit Resistivity			
Acquisition Dates:	07-Jan-2020		
Log Interval:	2629.00(m) – 2711.00(m)		
Index Types:	Measured Depth		
Index Scales:	1:200		
Depth Source:	Driller's Depth		
Depth Sensor:	3rd Party Depth		
Print Type:	Final		
Spud Date:	02-Dec-2019		
			

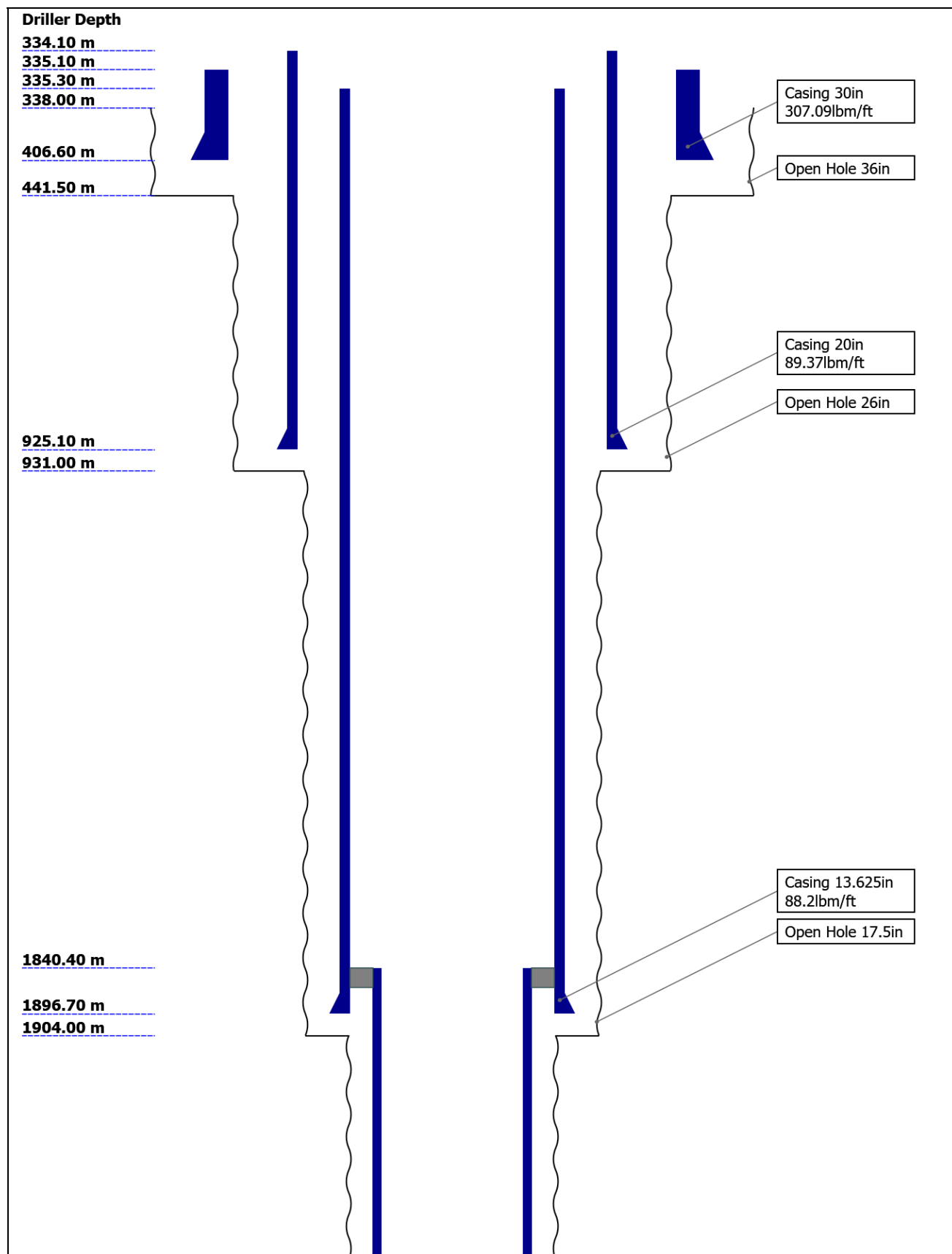
Disclaimer

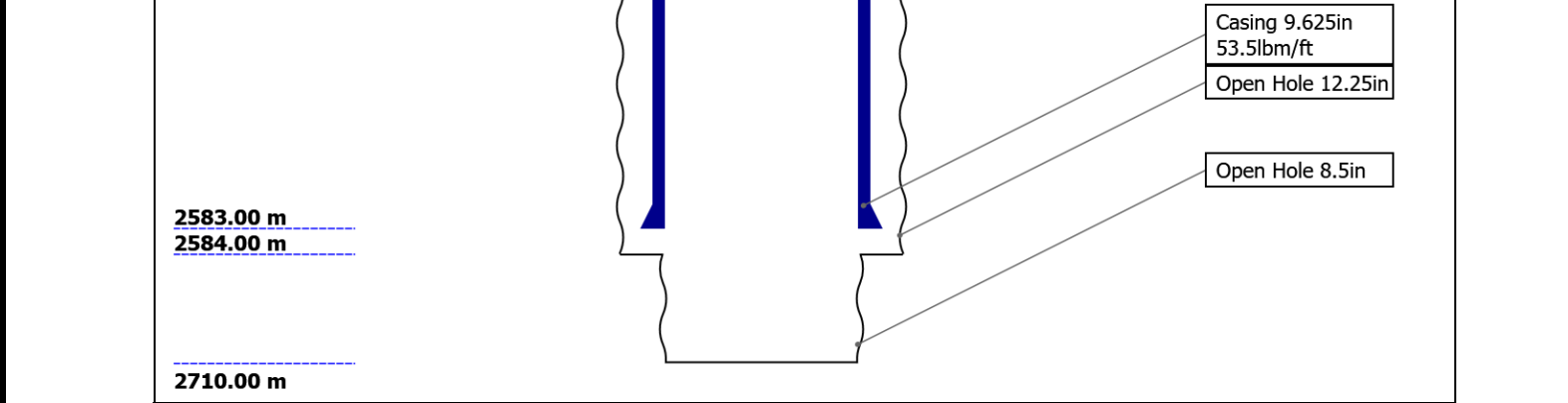
THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

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 - Integration Summary
 - Software Version
 - Composite Summary
 - Log (Resistivity - Dual Frequency)
 - Parameter Listing
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- Tail

Well Sketch





Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25	8.5	
Top Driller (m)	338	441.5	931	1904	2584	
Bottom Driller (m)	441.5	931	1904	2584	2710	
Casing						
Size (in)	30	20	13.625	9.625		
Weight (lbm/ft)	307.09	89.37	88.2	53.5		
Inner Diameter (in)	27.067	18.73	12.375	8.535		
Grade	X56	N80	P110	P110		
Top Driller (m)	335.1	334.1	335.3	1840.4		
Bottom Driller (m)	406.6	925.1	1896.7	2583		

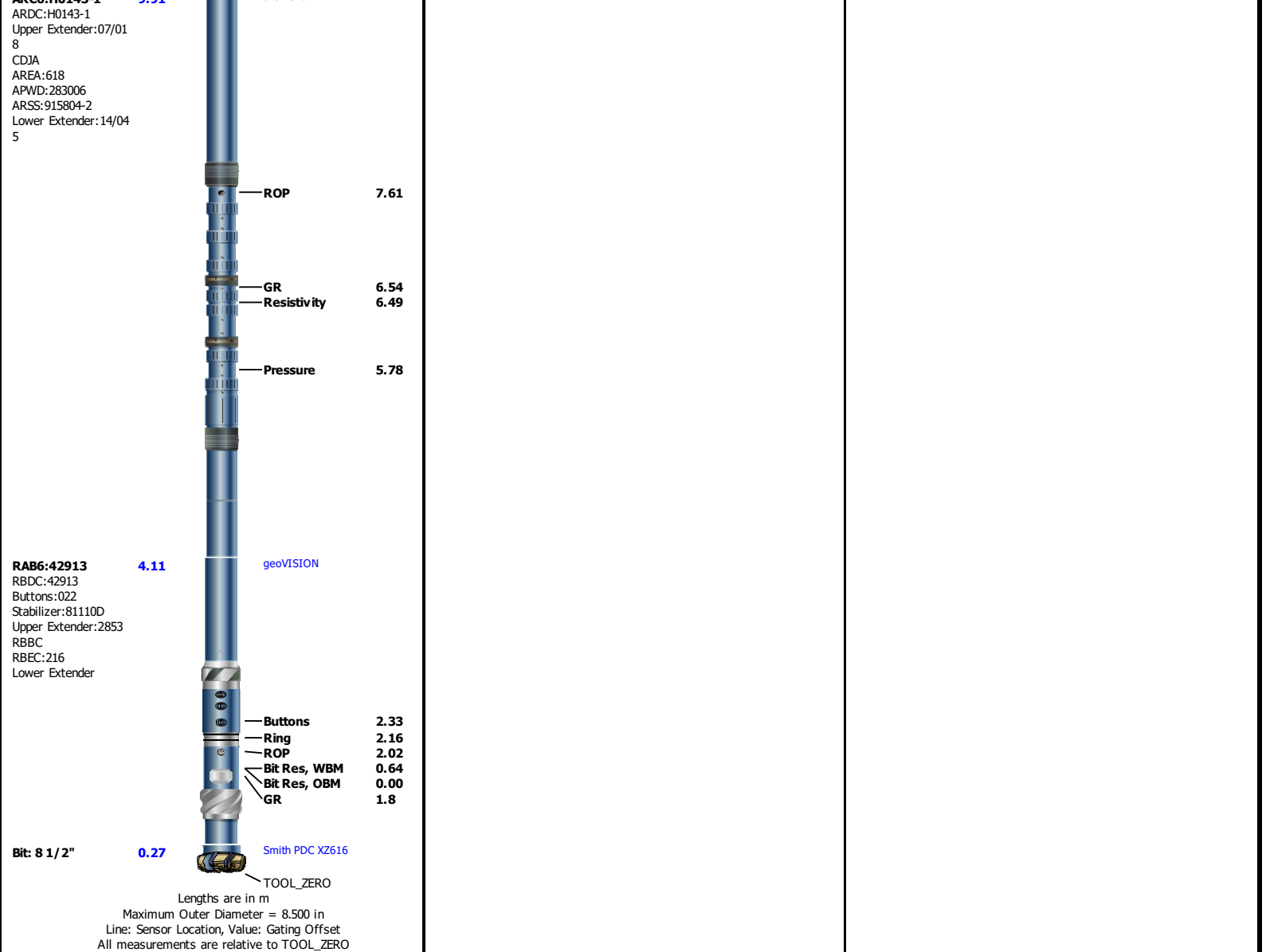
Operational Run Summary

Parameter (unit)	Run 12					
Date Log Started	07-Jan-2020					
Time Log Started	04:05:33					
Date Log Finished	07-Jan-2020					
Time Log Finished	17:16:13					
Bit Size (in)	8.500					
Bit Start Depth (m)	2697.00					
Bit Stop Depth (m)	2710.00					
Top Log Interval (m)	2630.90					
Bottom Log Interval (m)	2709.03					
Max Hole Deviation (deg)	1.12					
Azimuth of Max Deviation (deg)	120.28					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	M.Martter					
Witnessed By	R.Nordstokke					
Service Order Number	19SCA0085					

Borehole Fluids

Parameter(unit)	Run 12					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	44					
Source of Sample	Active Tank					
Salinity (ppm)	97140					
Density (g/cm3)	1.15					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.08 @ 14.8					
RMF @ Meas Temp (ohm.m@degC)	0.07 @ 14.8					
RMC @ Meas Temp (ohm.m@degC)	0.09 @ 20					
RM @ BHT (ohm.m@degC)	0.04 @ 44					
RMF @ BHT (ohm.m@degC)	0.04 @ 44					
RMC @ BHT (ohm.m@degC)	0.06 @ 44					
Total Solid (%)						
High Gravity Solids (%)						

Run 12: Toolstring			Run 12: Remarks	
Equip name	Length	MP name	Offset	All depths referenced to driller's depth. See EOWR for depth tracking details.
TELE675:E4388 MSSU675:30848 Upper Extender MDC675:E4388 MMA:2507 MDI:3666 PMVC:236 PMEA:803 MTA:2162 MTK675:004 MSSD675:1005482-1 Lower Extender, Long	18.37	TeleScope		All data from tool memory. All data acquired while drilling.
				Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.
				Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
				Data acquired while reaming down over cored interval from 2643 to 2697 m bit depth.
				Run 12 POOH at 2710 m to perform coring.
		D&I	14.14	
		Vibration	13.14	
		ROP	11.79	
ARC6-H0143-1	9.91	arcVISION		



Run 12

VISION Resistivity - Dual Frequency

Software Version

Acquisition System		Version	
Maxwell 2019.2		9.2.113335.3100	
Application Patch		DnM_TestKit-XceedCPU41-2019.2_9.2.114181	
Computation	Description	Version	
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family	9.2.113335.3100	
ARC6GammaRay	ARC6 Gamma Ray Computation Package for both Real-time and Recorded Mode	9.2.113335.3100	
GVRRes	GVR Resistivity Computation Package for both Real-time and Recorded Mode	9.2.113335.3100	
Tool Interface	System Version	Loaded Version	
HSPM	2019.0c.001	9.2.113335.3100	
Tool Elements	Description	Software Version	Firmware Version
RBEC	Electronics Chassis Assembly for RAB6-C	9.2.113335.3100	V8.8B
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 6.75 Inch Tool Drill Collar	9.2.113335.3100	V9.7B

Pass Summary

Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 12	Drilling	Down	298.98 m	2709.75 m	07-Jan-2020 4:05:33 AM	07-Jan-2020 5:16:13 PM	Yes

-|TICK_ARC_RES - Resistivity Tick Marks ARC6 RM

└ TICK_ARC_GR - Gamma Ray Tick Marks ARC6 RM

Attenuation Resistivity 40 inch
Spacing at 400 KHz,
Environmentally Corrected (A40L)
ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 40 inch
Spacing at 400 KHz,
Environmentally Corrected. (P40L)
ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 34 inch
Spacing at 400 KHz,
Environmentally Corrected. (P34L)
ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 28 inch
Spacing at 400 KHz,
Environmentally Corrected. (P28L)
ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 22 inch
Spacing at 400 KHz,
Environmentally Corrected. (P22L)
ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 16 inch
Spacing at 400 KHz,
Environmentally Corrected. (P16L)
ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 40 inch
Spacing at 2 MHz, Environmentally
Corrected. (P40H) ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 34 inch
Spacing at 2 MHz, Environmentally
Corrected. (P34H) ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 28 inch
Spacing at 2 MHz, Environmentally
Corrected. (P28H) ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 22 inch
Spacing at 2 MHz, Environmentally
Corrected. (P22H) ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Phase Shift Resistivity 16 inch
Spacing at 2 MHz, Environmentally
Corrected. (P16H) ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Bit Resistivity (RES_BIT)	RAB6	RM
0.2	ohm.m	200

[illegible]

Attenuation Resistivity 40 inch
Spacing at 2 MHz, Environmentally
Corrected. (A40H) ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Attenuation Resistivity 34 inch
Spacing at 2 MHz, Environmentally
Corrected (A34H) ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Attenuation Resistivity 28 inch
Spacing at 2 MHz, Environmentally
Corrected (A28H) ARC6 RM

0.2	ohm.m	200
Attenuation Resistivity 22 inch Spacing at 2 MHz - Environmentally		

Corrected (A22H) ARC6 RM

Attenuation Resistivity 16 inch
Spacing at 2 MHz, Environmentally
Corrected (A16H) ARC6 RM

0.2	ohm.m	200
-----	-------	-----

Resistivity Time After Bit (ARTM) ARC6

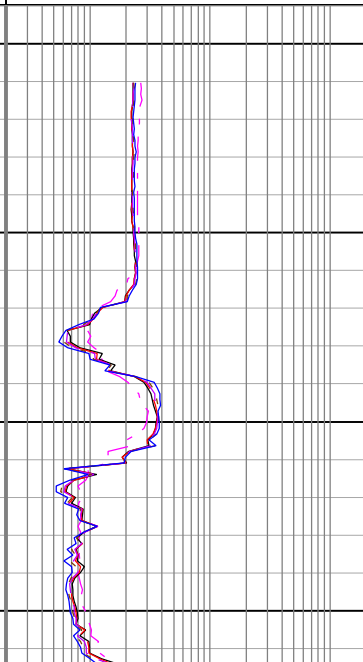
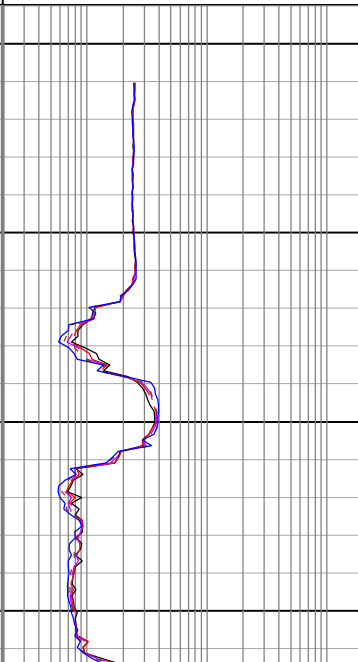
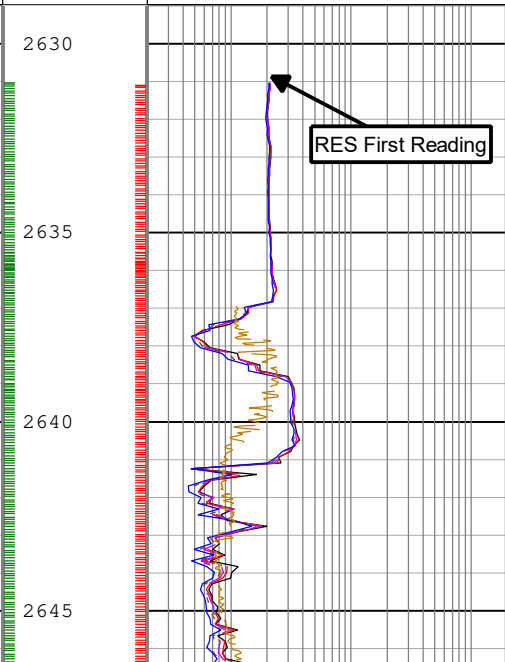
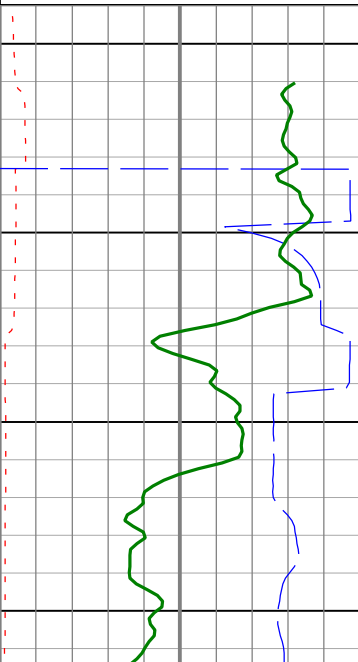
$0 \quad h \quad 10$

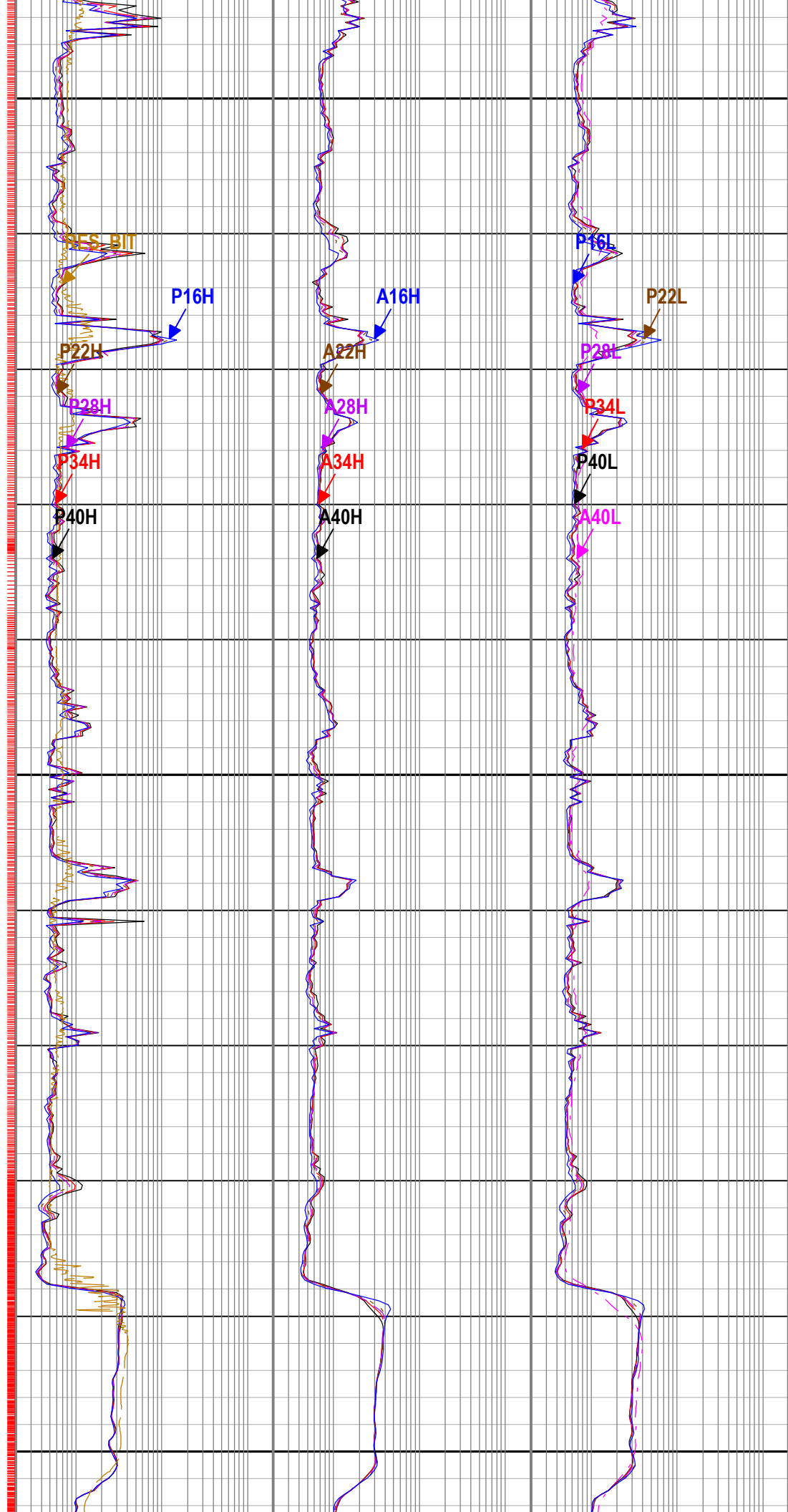
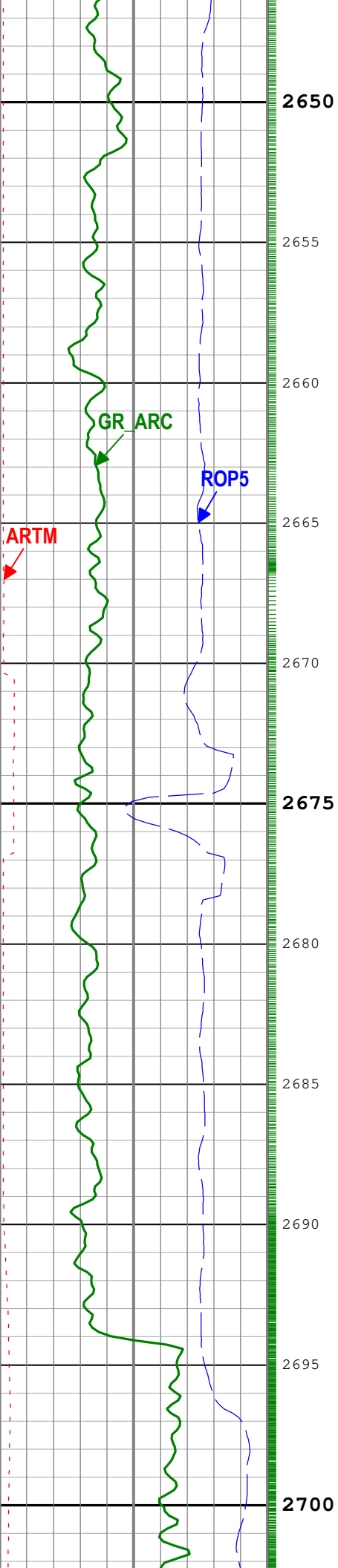
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT

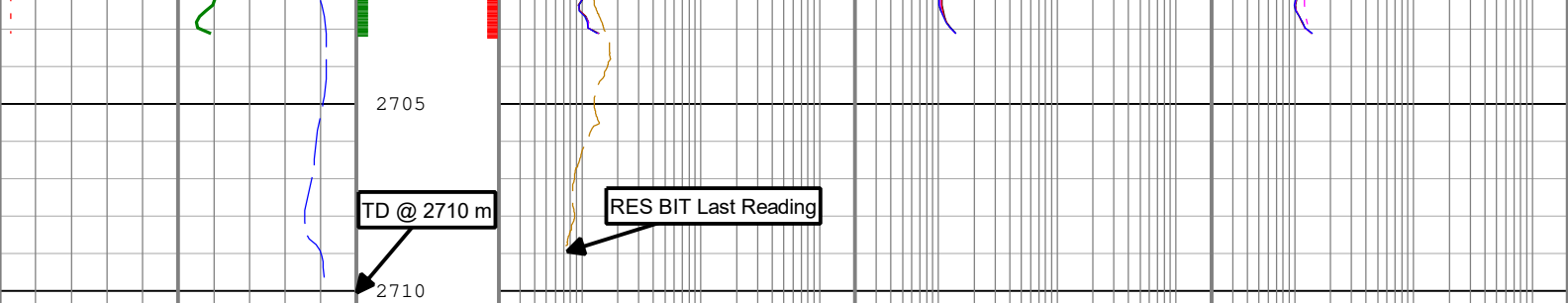
200 m/h 0

Gamma Ray (GR_ARC) ARC6 RM

0 gAPI 150







Resistivity Time After Bit (ARTM) ARC6		
0	h	10
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT		
200	m/h	0
Gamma Ray (GR_ARC) ARC6 RM		
0	gAPI	150

Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC6 RM		
0.2	ohm.m	200
Bit Resistivity (RES_BIT) RAB6 RM		
0.2	ohm.m	200

Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC6 RM		
0.2	ohm.m	200
Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (A34H) ARC6 RM		
0.2	ohm.m	200
Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (A28H) ARC6 RM		
0.2	ohm.m	200
Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (A22H) ARC6 RM		
0.2	ohm.m	200
Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (A16H) ARC6 RM		
0.2	ohm.m	200

Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (A40L) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC6 RM		
0.2	ohm.m	200
Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC6 RM		
0.2	ohm.m	200

└─ TICK_ARC_GR - Gamma Ray Tick Marks ARC6 RM

└─ TICK_ARC_RES - Resistivity Tick Marks ARC6 RM

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 20-Jan-2020 13:02:19

Channel Processing Parameters				
Run 12: Parameters				
Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC6	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	6	%
BHT	Bottom Hole Temperature	Borehole	44	degC
BS	Bit Size	DNMSESSION	8.5	in
DFD	Drilling Fluid Density	Borehole	1.15	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
GRSE_RM	Generalized Mud Resistivity Selection for Recorded Mode	Borehole	REMS(RM)	

GTSE_RM	Generalized Temperature Selection for Recorded Mode	Borehole	DHAT(RM)	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC6	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC6	1	ohm.m
MST	Mud Sample Temperature	Borehole	14.8	degC
RMS	Resistivity of Mud Sample	Borehole	0.08	ohm.m
TEMP_SEL_RAB	RAB Temperature Selection	RAB6	Tool	

Tool Control Parameters

Calibration Report

RAB6 (GeoVision Resistivity 675) Calibration - Run 12

Primary Equipment :

Electronics Chassis

RBEC

216

M21V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 2 at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

M22V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 2 at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250	

M01V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 0 at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9950	1.0250	

M02V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 0 at T2 Calibration Coefficient		Master	1.0000	0.9750	0.9980	1.0250	

R1V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Ring at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

R2V - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Ring at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250	

BDM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Deep at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0070	1.0250	

BDM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Deep at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0100	1.0250	

BMM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Medium at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9990	1.0250	

BMM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Medium at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250	

BSM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Shallow at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

BSM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Shallow at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250	

PGR - Gamma Ray: Blanket

Master (Time Frame File): 14:06:21 18-Dec-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.0000	0.7500	0.9838	1.2500	

ARC6 (Array Resistivity Compensated 675) Calibration - Run 12

Primary Equipment :

Elec. Chassis HP w/o AIM Receiver

AREA

618

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 15:26:50 07-Dec-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	7.951	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.997	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	4.584	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.904	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.142	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	1.927	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-1.871	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	1.878	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-1.879	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	1.856	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	7.970	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	6.989	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	4.593	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	4.889	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.160	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	-0.666	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	0.584	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	-0.629	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	0.582	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	-0.643	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:23:30 06-Jun-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.000	0.580	1.100	1.250	

Company: Equinor

Well: 31/5-7

Field: Eos
Rig Name: West Hercules
Country: Norway



VISION Resistivity
Dual Frequency
Recorded Mode, Run 12